

Home Work 7
II B. Tech. (Common)
Mathematics-II (NMCI102)

1. Examine whether the following matrices are diagonalizable or not. If so, obtain the matrix P such that $D = P^{-1}AP$ is a diagonal matrix.

(i) $A = \begin{bmatrix} 8 & -8 & -2 \\ 4 & -3 & -2 \\ 3 & -4 & 1 \end{bmatrix}$ (ii) $A = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 2 & 1 \\ -1 & 2 & 2 \end{bmatrix}$ (iii) $A = \begin{bmatrix} -17 & 18 & -6 \\ -18 & 19 & -6 \\ -9 & 9 & 2 \end{bmatrix}$ (iv) $A = \begin{bmatrix} 1 & -6 & -4 \\ 0 & 4 & 2 \\ 0 & -6 & -3 \end{bmatrix}$.

2. Let $A = \begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix}$. Obtain the modal matrix P and calculate the product $P^{-1}AP$. Also find A^{23} .
3. Find the matrix A , whose eigenvalues and the corresponding eigenvectors are as follows:
- (i) Eigenvalues: 1,2,3; Eigenvectors: $(1, 2, 1)^T$, $(2, 3, 4)^T$, $(1, 4, 9)^T$.
- (ii) Eigenvalues: 0,0,3; Eigenvectors: $(1, 2, -1)^T$, $(-2, 1, 0)^T$, $(3, 0, 1)^T$.
- (iii) Eigenvalues: 20,18,0,0; Eigenvectors: $(3, -3, -5, 5)^T$, $(1, 1, 9, 9)^T$, $(0, 1, 0, 0)^T$, $(1, 0, 0, 0)^T$.
4. Determine diagonal matrices orthogonally similar to the following real symmetric matrices, obtaining also the transforming matrices:

(i) $\begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$ (ii) $\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$ (iii) $\begin{bmatrix} 7 & 4 & -4 \\ 4 & -8 & -1 \\ -4 & -1 & -8 \end{bmatrix}$ (iv) $\begin{bmatrix} 7 & 0 & -2 \\ 0 & 5 & -2 \\ -2 & -2 & 6 \end{bmatrix}$.

5. Write down the quadratic forms corresponding to the following symmetric matrices:

(i) $\begin{bmatrix} 2 & -3 & 1 \\ -3 & 2 & 4 \\ 1 & 4 & -5 \end{bmatrix}$ (ii) $\begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}$ (iii) $\begin{bmatrix} 1 & -1 & 0 & \frac{3}{2} \\ -1 & -2 & 2 & 0 \\ 0 & 2 & 4 & -\frac{5}{2} \\ \frac{3}{2} & 0 & -\frac{5}{2} & -4 \end{bmatrix}$ (iv) $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 2 & \frac{5}{2} \\ 3 & \frac{5}{2} & 3 \end{bmatrix}$.

6. Reduce the following quadratic form to the canonical form by an orthogonal transformation. Also specify the matrix of transformation in each case and determine the rank, nature, index and signature.

(i) $8x^2 + 7y^2 + 3z^2 - 12xy - 8yz + 4zx$ (ii) $6x^2 + 3y^2 + 14z^2 + 4xy + 4yz + 18zx$.

7. Find the eigenvalues and eigenvectors of the matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & -1 \\ 0 & -1 & 3 \end{bmatrix}$ and hence reduce the quadratic

form $Q = x_1^2 + 3x_2^2 + 3x_3^2 - 2x_2x_3$ to canonical form. Also determine the rank, nature, index and signature.

8. Find out what type of conic section the quadratic equation $17x_1^2 - x_1x_2 + 17x_2^2 = 128$ represents and transform it to principal axes.

9. Show that

- (i) The eigenvalues of a Hermitian matrix or a real symmetric matrix are real.
- (ii) The eigenvalues of a skew-Hermitian matrix are zero or pure imaginary.
- (iii) The eigenvalues of a unitary matrix are of magnitude 1.
- (iv) The eigenvalues of an idempotent matrix ($A^2 = A$) are either 0 or 1.
- (v) The eigenvalues of a nilpotent matrix ($A^k = 0$) are all zero.
- (vi) Eigenvectors corresponding to distinct eigenvalues of a matrix are linearly independent.
- (vii) Eigenvectors corresponding to distinct eigenvalues of a real symmetric matrix are orthogonal.
- (viii) For any real $m \times n$ matrix A , the matrices $A^T A$ and AA^T are symmetric with non-negative eigenvalues and share the same non-zero eigenvalues.